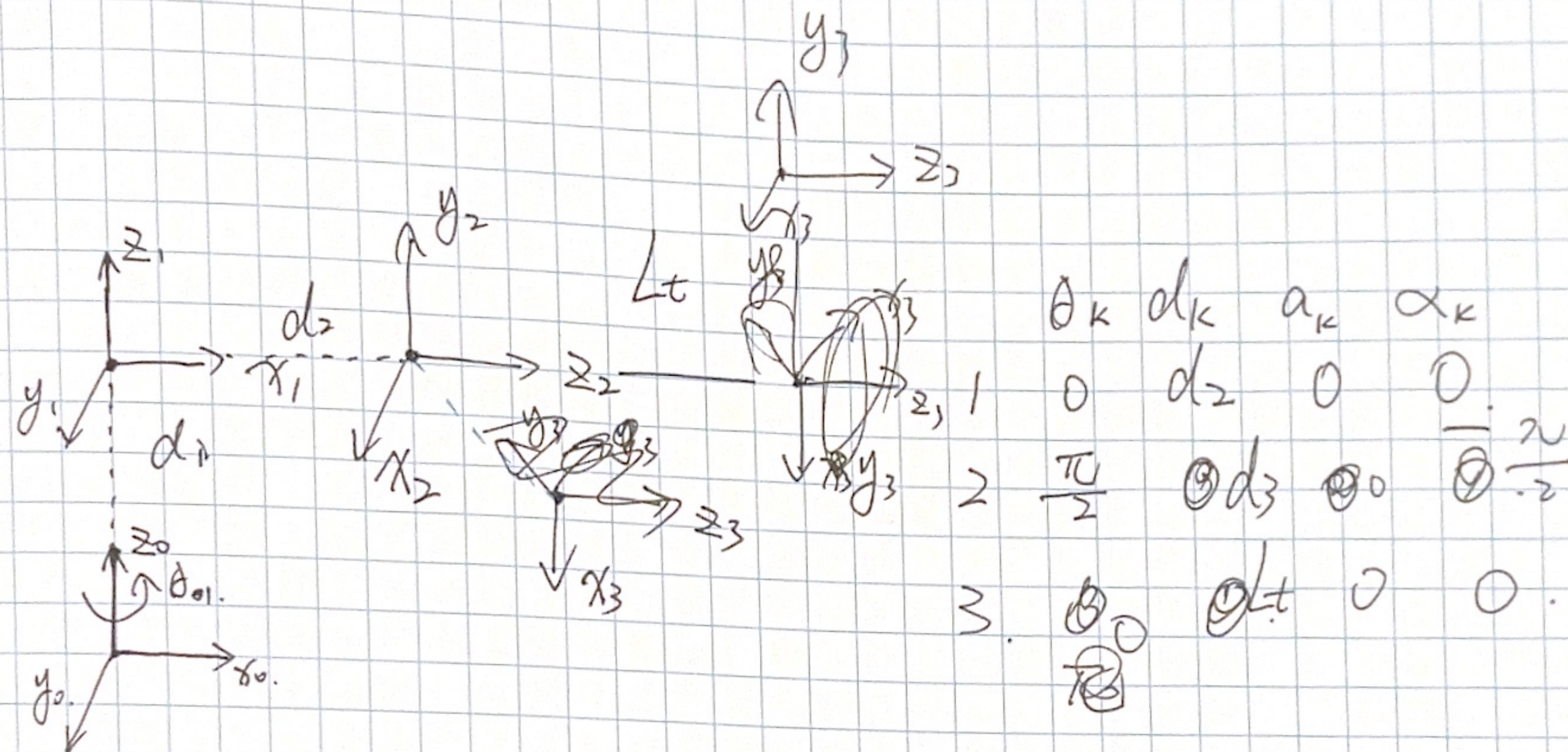


9. path is the straight line in-between 2 pos, no time involved but



$$3. {}^0T_1 = \begin{bmatrix} I & \begin{matrix} d_2 \\ 0 \\ 0 \end{matrix} \\ 0 & 1 \end{bmatrix}$$

$${}^1T_2 = R_z\left(\frac{\pi}{2}\right) \text{Trans}_z(d_3) \left(R_x\left(-\frac{\pi}{2}\right) \right)$$

$${}^2T_3 = \text{Trans}_z(L_t)$$

$$4. {}^0T_{\text{tool}} = {}^0T_1 {}^1T_2 {}^2T_3$$

Q2.

$${}^0T_{pick} = \begin{bmatrix} 0 & 1 & 0 & 4 \\ 1 & 0 & 0 & 8 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$b. {}^0T_{place} = \begin{bmatrix} 0 & 1 & 0 & 9 \\ 1 & 0 & 0 & 3 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3 year.

$$C. \begin{bmatrix} (d_3 + L_1) \cos \theta_1 \\ (d_3 + L_1) \sin \theta_1 \\ d_2 \\ e^{\theta_1/\pi} \cos \theta_1 \\ e^{\theta_1/\pi} \sin \theta_1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} x &= (d_3 + L_1) \cos \theta_1 \\ y &= (d_3 + L_1) \sin \theta_1 \\ z &= d_2 \\ \theta_1 &= \text{atan2}(y, x) \end{aligned}$$

$$\begin{cases} \theta_1 = \text{atan2}(y, x) \\ d_2 = z \\ d_3 = \sqrt{x^2 + y^2} - L_1 \end{cases}$$